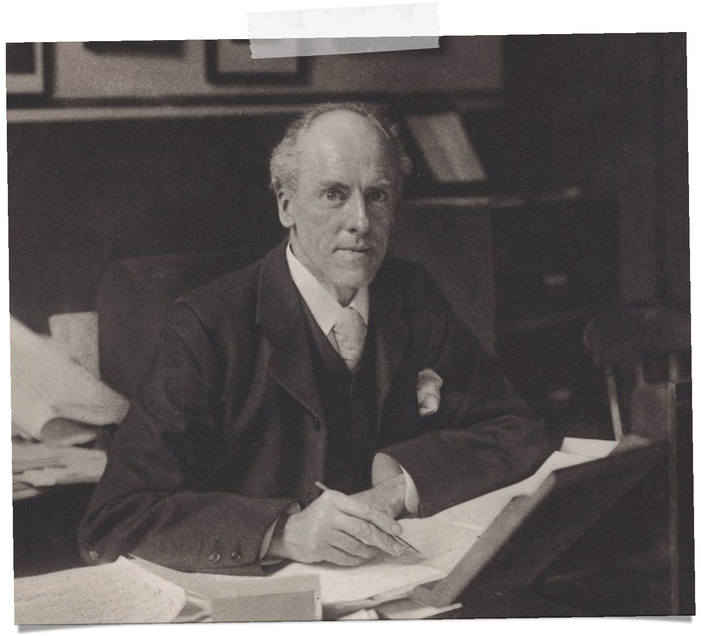
Chi-Square (χ^2) Test

2026. 10. 29.

Based in part on lecture notes by Prof. Byoungkwan Lee and 이병관 (2023). 사회과학 연구자를 위한 R을 활용한 기초 통계. 한양대학교출판부.

History of the Chi-Square Test



Karl Pearson
(March 1857 – April 1936)

관찰된 데이터와 이론적으로 기대되는 데이터 사이의 차이가
우연한 표본추출로 설명될 수 있을까?

Image source: https://en.wikipedia.org/wiki/Karl_Pearson

History of the Chi-Square Test

The following data are due to Professor W. F. R. Weldon, F.R.S., and give the observed frequency of dice with 5 or 6 points when a cast of twelve dice was made 26,306 times :—

No. of Dice in Cast with 5 or 6 Points.	Observed Frequency, m'.	Theoretical Frequency, m.	Deviation, e.
0	185	203	− 18
1	1149	1217	− 68
2	3265	3345	− 80
3	5475	5576	−101
4	6114	6273	−159
5	5194	5018	+176
6	3067	2927	+140
7	1331	1254	+ 77
8	403	392	+ 11
9	105	87	+ 18
10	14	13	+ 1
11	4	1	+ 3
12	0	0	0
	26306	26306	

The results show a bias from the theoretical results, 5 and 6 points occurring more frequently than they should do. Are the deviations such as to forbid us to suppose the results due to random selection? Is there in apparently true dice a real bias towards those faces with the maximum number of points appearing uppermost?

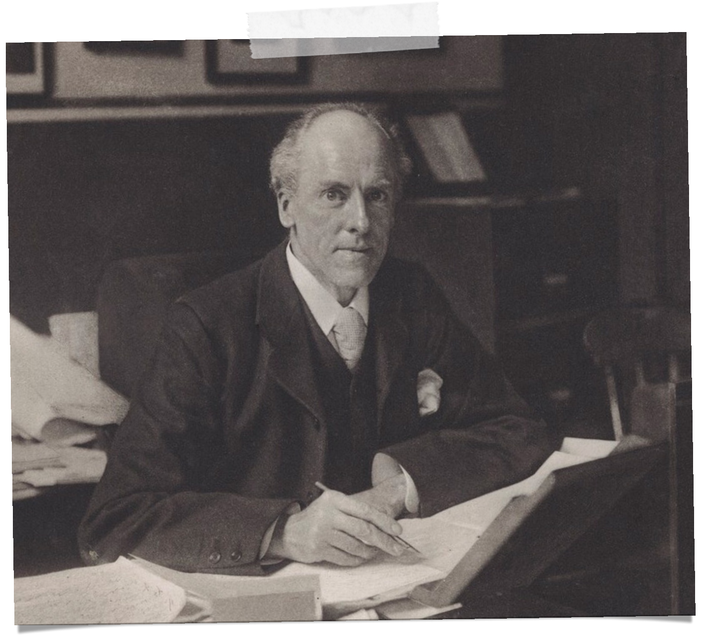
공정한(fair) 주사위라면 5 또는 6이 나올 확률은 1/3..

그런데 왜 더 자주 나올까?

"실제 관찰된 빈도가 이론적으로 기대되는 빈도와 다른 이유가 우연 때문인가?"

Pearson, K. (1900). X. On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science, 50*(302), 157-175.

History of the Chi-Square Test



Karl Pearson
(March 1857 – April 1936)

Hence $\chi^2 = 43{\cdot}87241$ and $\chi = 6{\cdot}623{,}625$.

As there are 13 groups we have to find P from the formula :

$$P = e^{-\frac{1}{2}\chi^2}\left(1 + \frac{\chi^2}{2} + \frac{\chi^4}{2.4} + \frac{\chi^6}{2.4.6} + \frac{\chi^8}{2.4.6.8} + \frac{\chi^{10}}{2.4.6.8.10}\right),$$

which leads us to

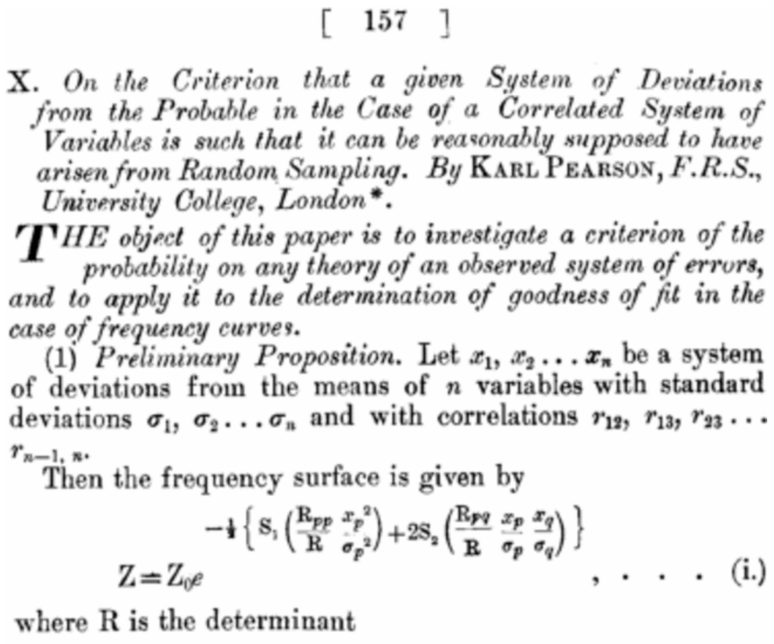
$$P = {\cdot}000016,$$

or the odds are 62,499 to 1 against such a system of deviations on a random selection. With such odds it would be reasonable to conclude that dice exhibit bias towards the higher points.

The dice are unfair !!

Pearson, K. (1900). X. On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science, 50*(302), 157-175.

History of the Chi-Square Test



[157]

X. *On the Criterion that a given System of Deviations from the Probable in the Case of a Correlated System of Variables is such that it can be reasonably supposed to have arisen from Random Sampling.* By KARL PEARSON, *F.R.S., University College, London**.

THE object of this paper is to investigate a criterion of the probability on any theory of an observed system of errors, and to apply it to the determination of goodness of fit in the case of frequency curves.

(1) *Preliminary Proposition.* Let $x_1, x_2 \ldots x_n$ be a system of deviations from the means of n variables with standard deviations $\sigma_1, \sigma_2 \ldots \sigma_n$ and with correlations $r_{12}, r_{13}, r_{23} \ldots r_{n-1,n}$.

Then the frequency surface is given by

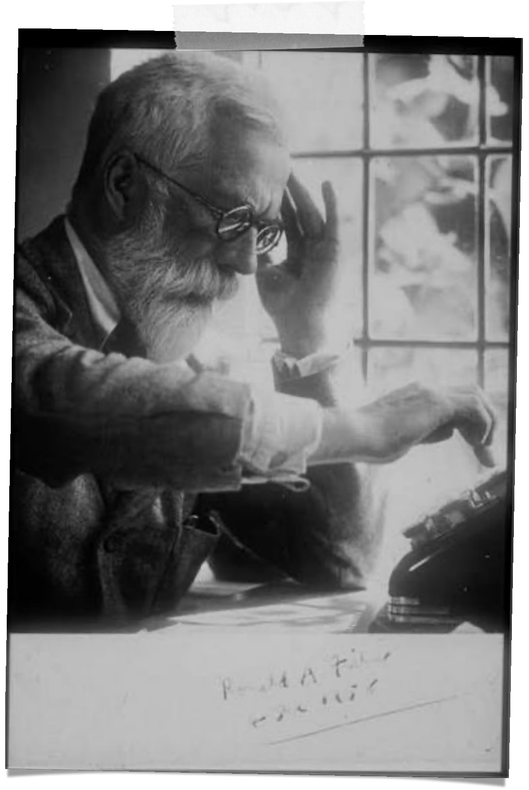
$$Z = Z_0 e^{-\frac{1}{2}\left\{S_1\left(\frac{R_{pp}}{R}\frac{x_p^2}{\sigma_p^2}\right)+2S_2\left(\frac{R_{pq}}{R}\frac{x_p}{\sigma_p}\frac{x_q}{\sigma_q}\right)\right\}} \quad \ldots \quad \text{(i.)}$$

where R is the determinant

- "상관관계가 있는 변수 체계의 경우, 주어진 편차 시스템이 무작위 샘플링에서 발생했다고 합리적으로 추정할 수 있는 기준에 대하여"
- The χ^2-distribution was developed earlier, but Pearson (1900) was the first to provide a table of p-values corresponding to χ^2-values.
- Therefore, the χ^2-test is also known as **Pearson's χ^2-test** (Gorroochurn, 2016).

Pearson, K. (1900). X. On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science, 50*(302), 157-175.
Gorroochurn, P. (2016). Classic topics on the history of modern mathematical statistics: From Laplace to more recent times. John Wiley & Sons.

History of the Chi-Square Test



Sir Ronald Aylmer Fisher
(February 1890 – July 1962)

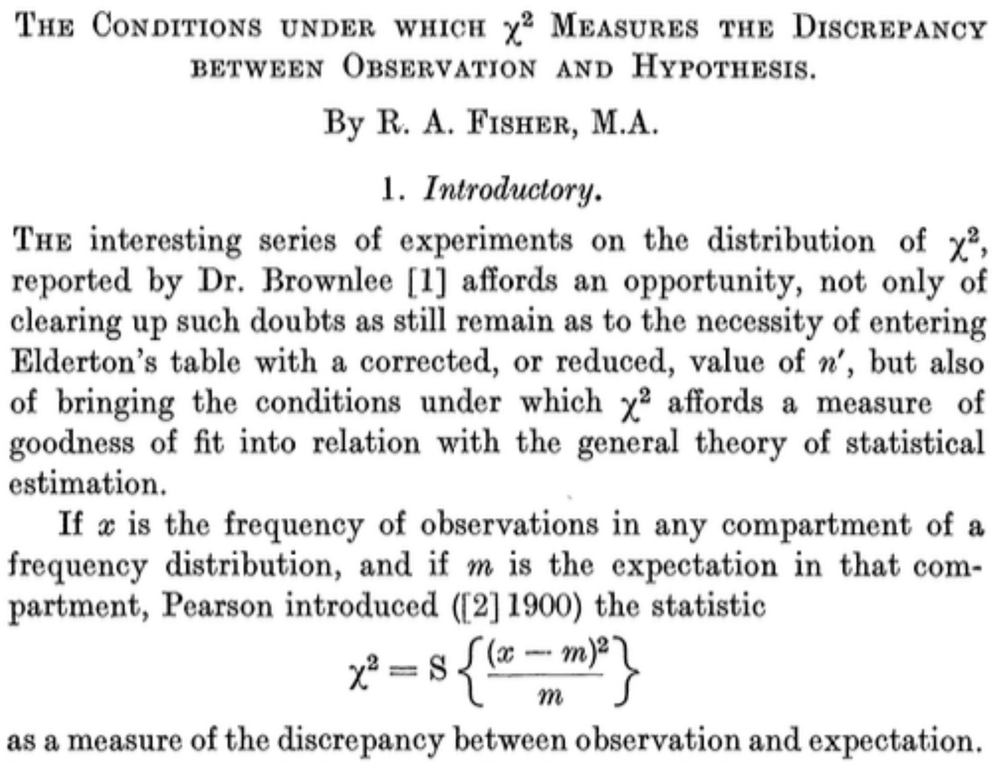
"공정한(fair) 주사위라면 5 또는 6이 나올 확률은 1/3" 이라고?

이 확률은 표본에서 추정한 게 아니라, 이론적으로 이미 알고 있던 거잖아?

표본에서 모수를 추정할 경우 카이제곱 통계량이 자동으로 원래의 카이제곱 분포를 따르는 것은 아니지 않을까?

Image source: https://www.nature.com/articles/s41437-020-00394-6

History of the Chi-Square Test



THE CONDITIONS UNDER WHICH χ^2 MEASURES THE DISCREPANCY BETWEEN OBSERVATION AND HYPOTHESIS.

By R. A. FISHER, M.A.

1. *Introductory.*

THE interesting series of experiments on the distribution of χ^2, reported by Dr. Brownlee [1] affords an opportunity, not only of clearing up such doubts as still remain as to the necessity of entering Elderton's table with a corrected, or reduced, value of n', but also of bringing the conditions under which χ^2 affords a measure of goodness of fit into relation with the general theory of statistical estimation.

If x is the frequency of observations in any compartment of a frequency distribution, and if m is the expectation in that compartment, Pearson introduced ([2] 1900) the statistic

$$\chi^2 = \mathrm{S}\left\{\frac{(x-m)^2}{m}\right\}$$

as a measure of the discrepancy between observation and expectation.

- Pearson introduced the χ^2-test, but Fisher (1924) clarified its theoretical conditions.
- If parameters are **estimated from the sample**, the reference distribution is affected.
- Therefore, **the degrees of freedom must be adjusted.**

Fisher, R. A. (1924). The conditions under which χ2 measures the discrepancey between observation and hypothesis. *Journal of the Royal Statistical Society*, 442-450.

History of the Chi-Square Test

Why Can χ^2 Be Abnormally Distributed? (Fisher, 1924)

- False hypothesis (If the hypothesis tested is not in fact true.)
- Inconsistent estimation (If the method of estimation employed is Inconsistent.)
- Inefficient estimation (If the method of estimation employed is Inefficient.)

Fisher's key contribution: Df must be reduced to account for parameters estimated from the sample.

→ Thus, today, we apply Fisher's df correction when estimating unknown parameters in Pearson's chi-square test.

Fisher, R. A. (1924). The conditions under which χ2 measures the discrepancey between observation and hypothesis. *Journal of the Royal Statistical Society*, 442-450.

Parametric vs. Nonparametric Tests

Assumption of parametric statistics

- normality
- homogeneity of variance
- independence

Parametric vs. Nonparametric Tests

When choosing the type of statistical method (parametric vs. non-parametric), several factors should be considered:

- Scale of measurement of the data, e.g., nominal, ordinal, interval, ratio, continuous, discrete, etc.
- Distribution of the population, e.g., normal versus non-normal
- Homogeneity of variances, e.g., equal versus unequal variances
- Independence versus dependency of the drawn samples considering the variables.

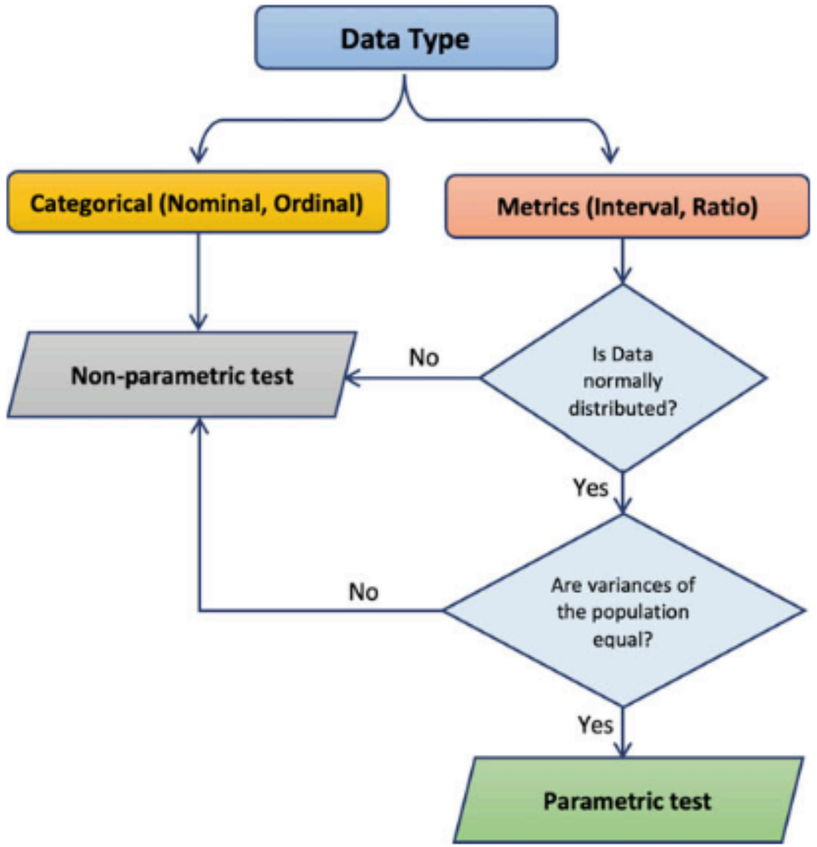


Fig. 4.3 Choosing between parametric and non-parametric tests

Okoye, K., & Hosseini, S. (2024). Choosing between parametric and non-parametric tests in statistical data analysis. In *R programming: Statistical data analysis in research* (pp. 87-98). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-97-3385-9_4

Parametric vs. Nonparametric Tests

- Previous examples of hypothesis tests, such as the *t*-tests are **parametric tests** and they do include assumptions about parameters and hypotheses about parameters.

- The term “non-parametric” refers to the fact that the chi-square tests do not require assumptions about population parameters nor do they test hypotheses about population parameters.

- The most obvious difference between the chi-square tests and the other hypothesis tests we have considered (*t* and ANOVA) is the nature of the data.

- For chi-square, the data are frequencies rather than numerical scores.

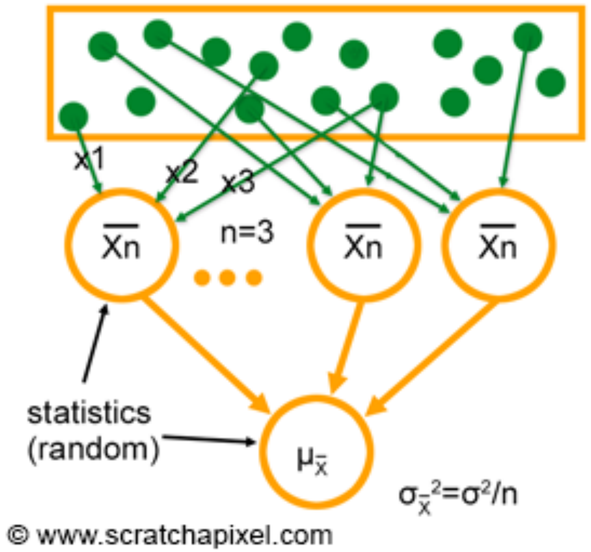
Parametric vs. Nonparametric Tests

	Parametric	Non-parametric
Assumed distribution	Normal	Any
Assumed variance	Homogeneous	Any
Typical data	Ratio or Interval	Ordinal or Nominal
Data set relationships	Independent	Any
Usual central measure	Mean	Median
Benefits	Can draw more conclusions	Simplicity; Less affected by outliers
Tests		
Choosing	Choosing parametric test	Choosing a non-parametric test
Correlation test	Pearson	Spearman
Independent measures, 2 groups	Independent-measures t-test	Mann-Whitney test
Independent measures, >2 groups	One-way, independent-measures ANOVA	Kruskal-Wallis test
Repeated measures, 2 conditions	Matched-pair t-test	Wilcoxon test
Repeated measures, >2 conditions	One-way, repeated measures ANOVA	Friedman's test

Why use a Chi-square statistic?

- Most technique we examine works for interval and ratio types of data
- Correlation even works pretty well for ordinal data
- What about nominal or categorical data?

Quick review



population
μ: population mean
σ^2: population variance

$\overline{X}_n$: sample mean
S_n: sample variance

$\mu_{\bar{x}}$: mean (expected value) of sample mean
$\sigma_{\bar{x}}^2$: variance of sample mean

VARIANCE FORMULAS: POPULATION vs. SAMPLE

POPULATION VARIANCE

$$\sigma^2 = \frac{1}{n}\sum(x_i - \mu)^2$$

Divisor: Total Population Size

σ^2 = Population Variance
n = Population Size
x_i = Individual Data Points
μ = Population Mean

SAMPLE VARIANCE

$$s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$$

Why n-1? (Bessel's Correction)

Divisor: Degrees of Freedom (Unbiased Estimator)

s^2 = Sample Variance
n = Sample Size
x_i = Individual Data Points
$\bar{x}$ = Sample Mean

The use of "n-1" in the sample variance formula corrects for the bias introduced when estimating the population variance from a sample, ensuring the result is on average equal to the true population variance.

Image source: https://www.scratchapixel.com/lessons/mathematics-physics-for-computer-graphics/monte-carlo-methods-mathematical-foundations//estimators.html, https://medium.com/@tomkob99_89317/what-is-this-evil-1-in-sample-variance-0427bb1de598

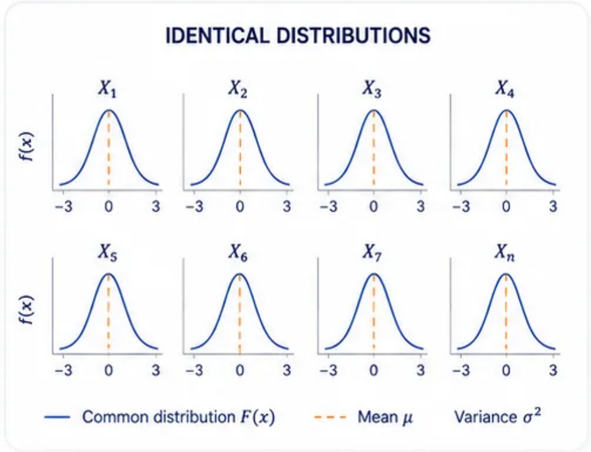
Quick review

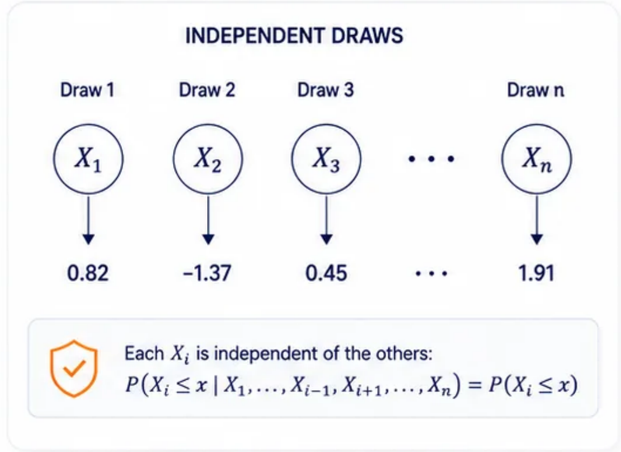
Sampling Distribution of Sample Mean

If the population distribution is $N(\mu, \sigma^2)$:

- $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$
- $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$
- $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$







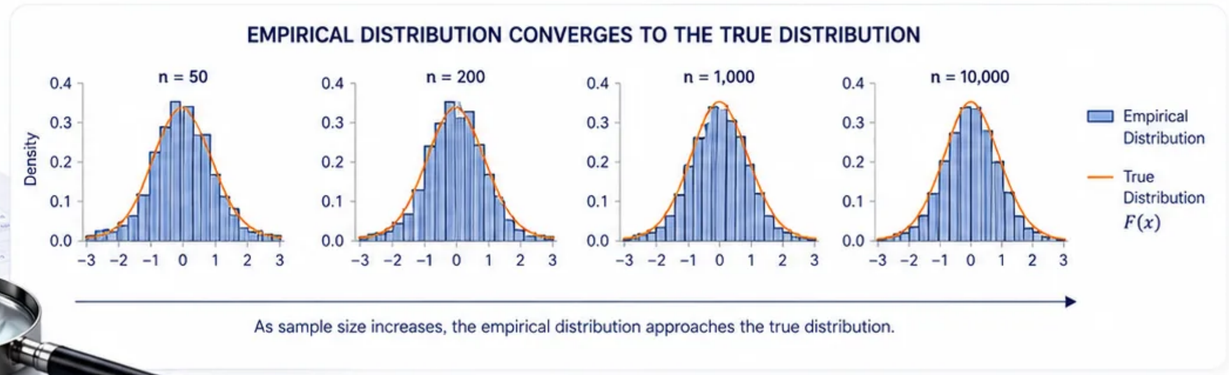


Image source: https://riskhub.org/blogs/iid-random-variables-explained

χ^2-distribution

- $Z_i \sim N(0,1)$
- $Z_1, Z_2, \ldots, Z_\nu \overset{i.i.d.}{\sim} N(0,1)$
- $\chi^2 = Z_1^2 + Z_2^2 + \cdots + Z_\nu^2 = \sum_{i=1}^{\nu} Z_i^2 \sim \chi^2(\nu)$

χ^2-distribution

Sampling Distribution of Sample Variance

If the population distribution is $N(\mu, \sigma^2)$:

- $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$
- A chi-squared distribution is a continuous distribution with ν degrees of freedom.
- It is used to describe the distribution of a sum of squared random variables.

χ^2-distribution

Derive Sampling Distribution of Sample Variance

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{X})^2}{n-1}$$

$$\frac{(n-1)s^2}{\sigma^2} = \frac{\sum_{i=1}^{n}(x_i - \bar{X})^2}{\sigma^2}$$

$$\frac{(n-1)s^2}{\sigma^2} = \sum_{i=1}^{n}\left(\frac{x_i - \bar{X}}{\sigma}\right)^2$$

$$\frac{(n-1)s^2}{\sigma^2} = \sum_{i=1}^{n} Z_i^2$$

$$\frac{(n-1)s^2}{\sigma^2} = \sum_{i=1}^{n} Z_i^2 \sim \chi^2(n-1)$$

Cf. why n−1?

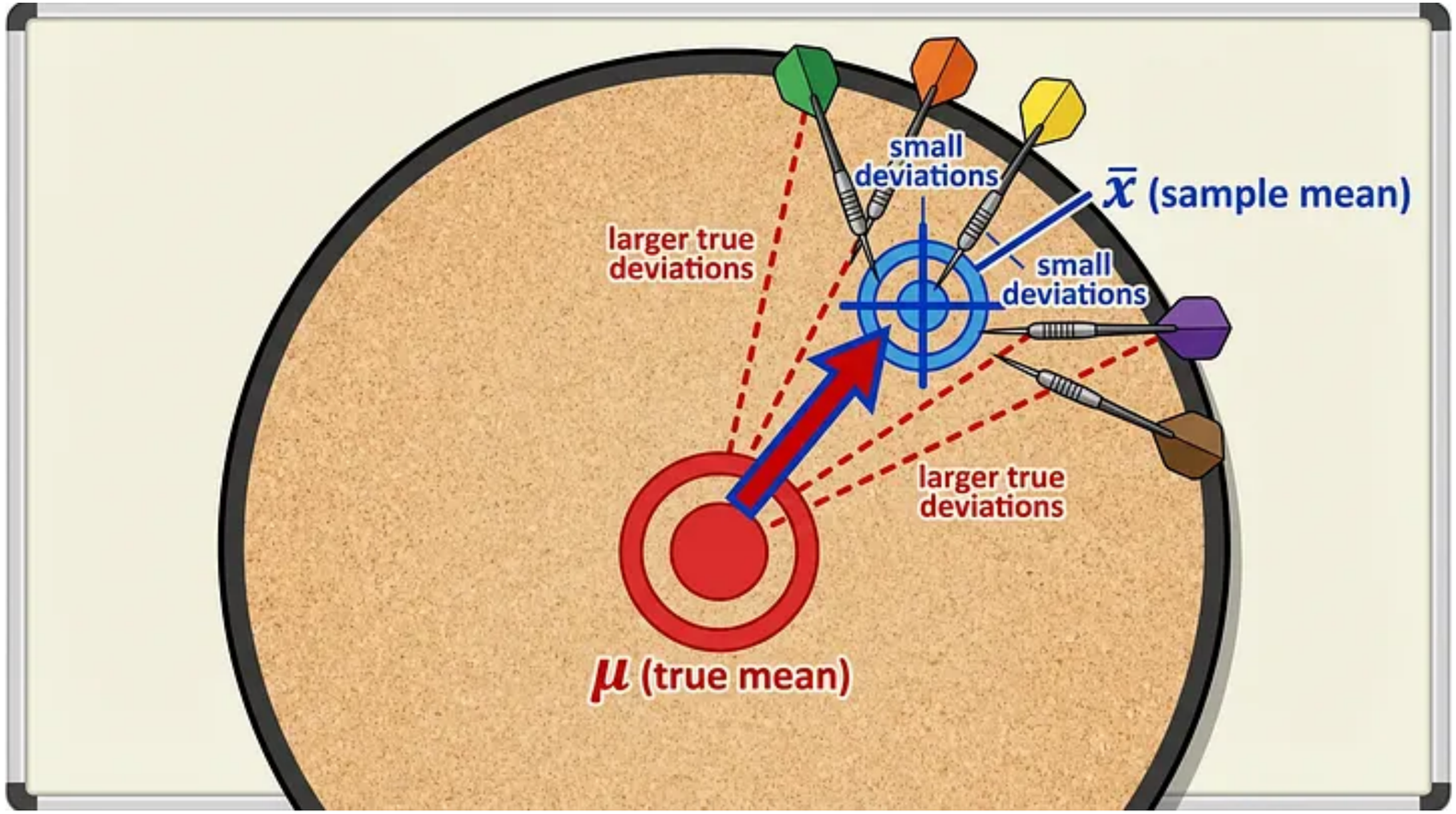


Image & explanation source: https://medium.com/@tomkob99_89317/what-is-this-evil-1-in-sample-variance-0427bb1de598

Cf. why n-1?

- Small samples: The bias is severe. Using n dramatically underestimates variance.
- Large samples: As n grows, both estimators converge toward the true variance.
- The correction: Using $n-1$ provides an unbiased estimate across all sample sizes.

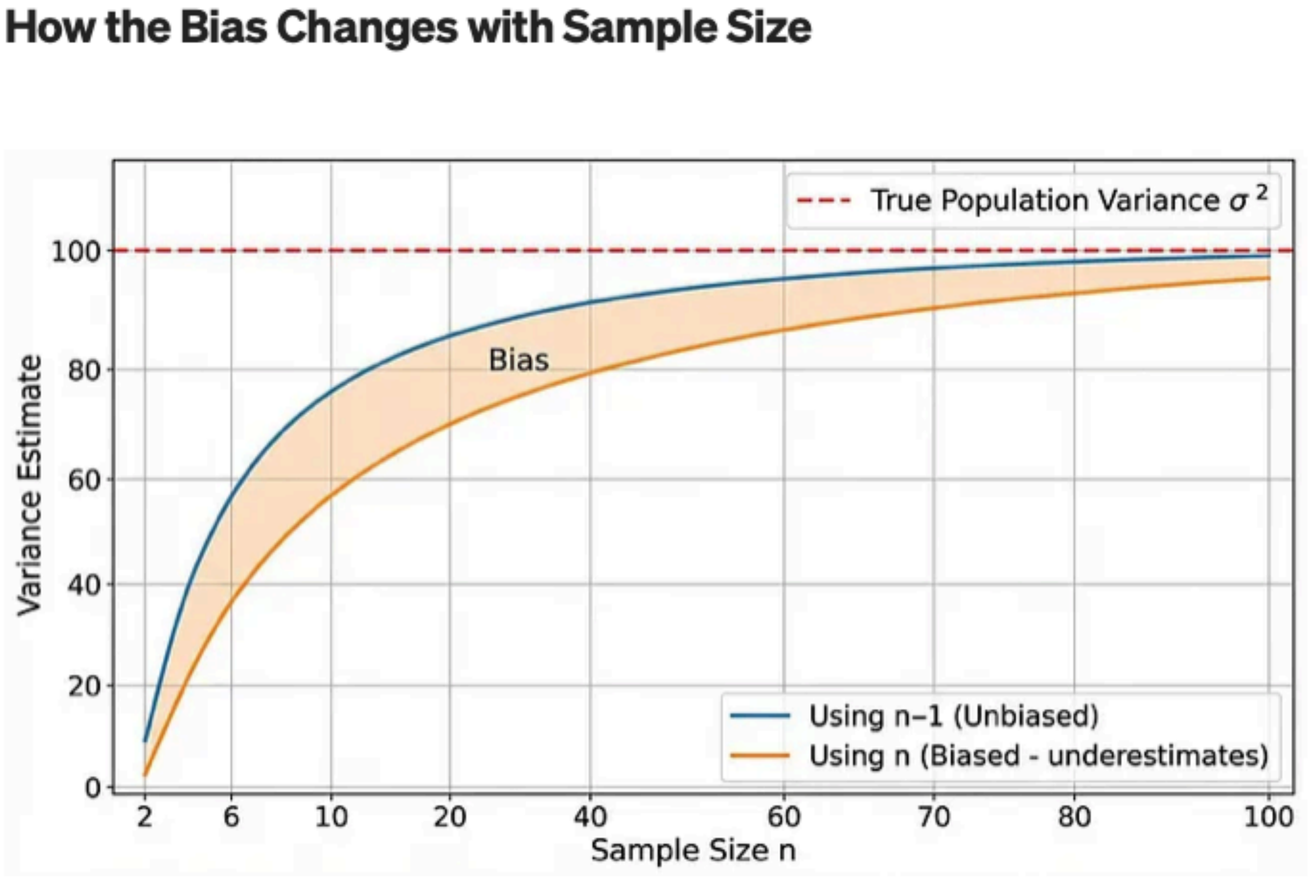
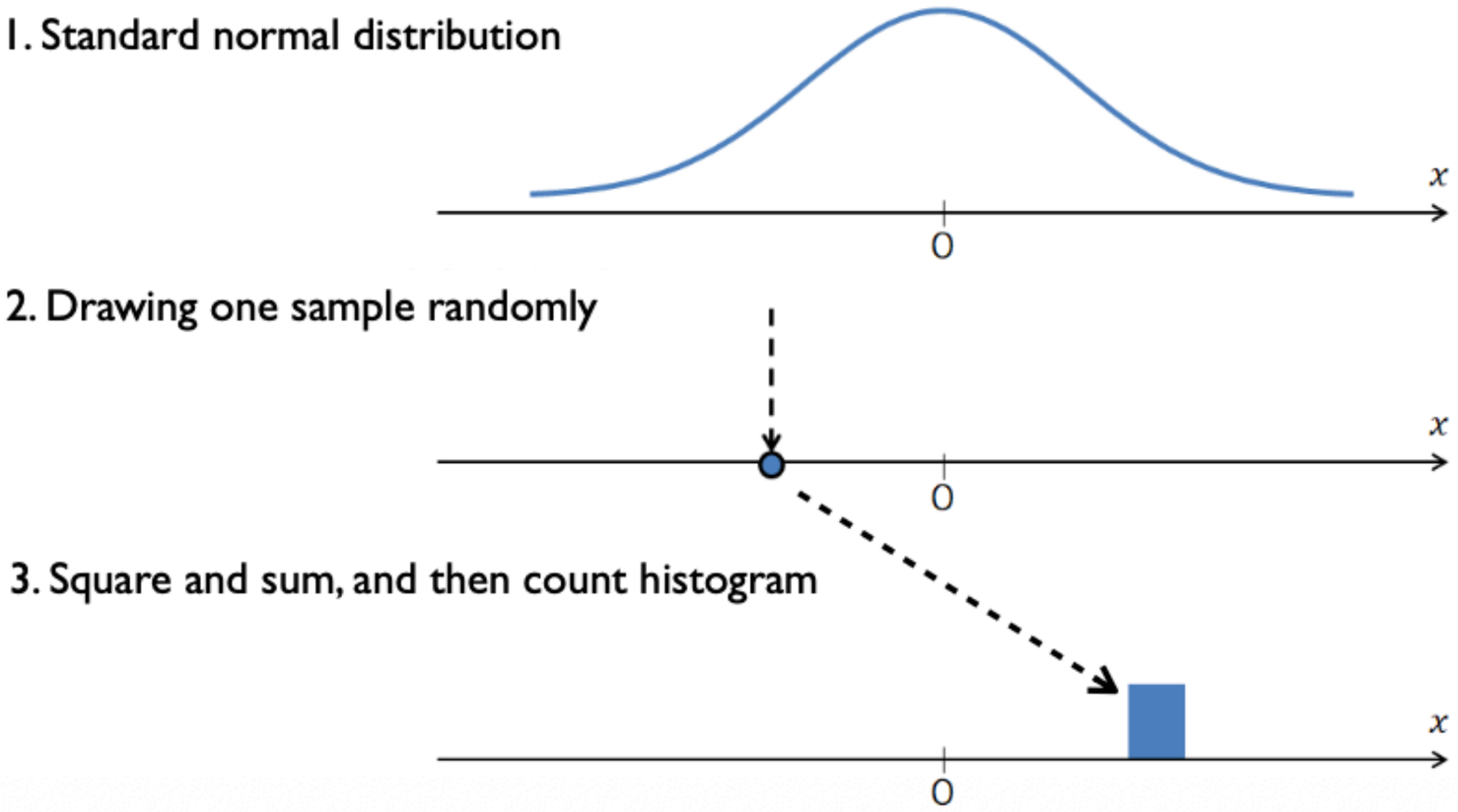


Image & explanation source: https://medium.com/@tomkob99_89317/what-is-this-evil-1-in-sample-variance-0427bb1de598

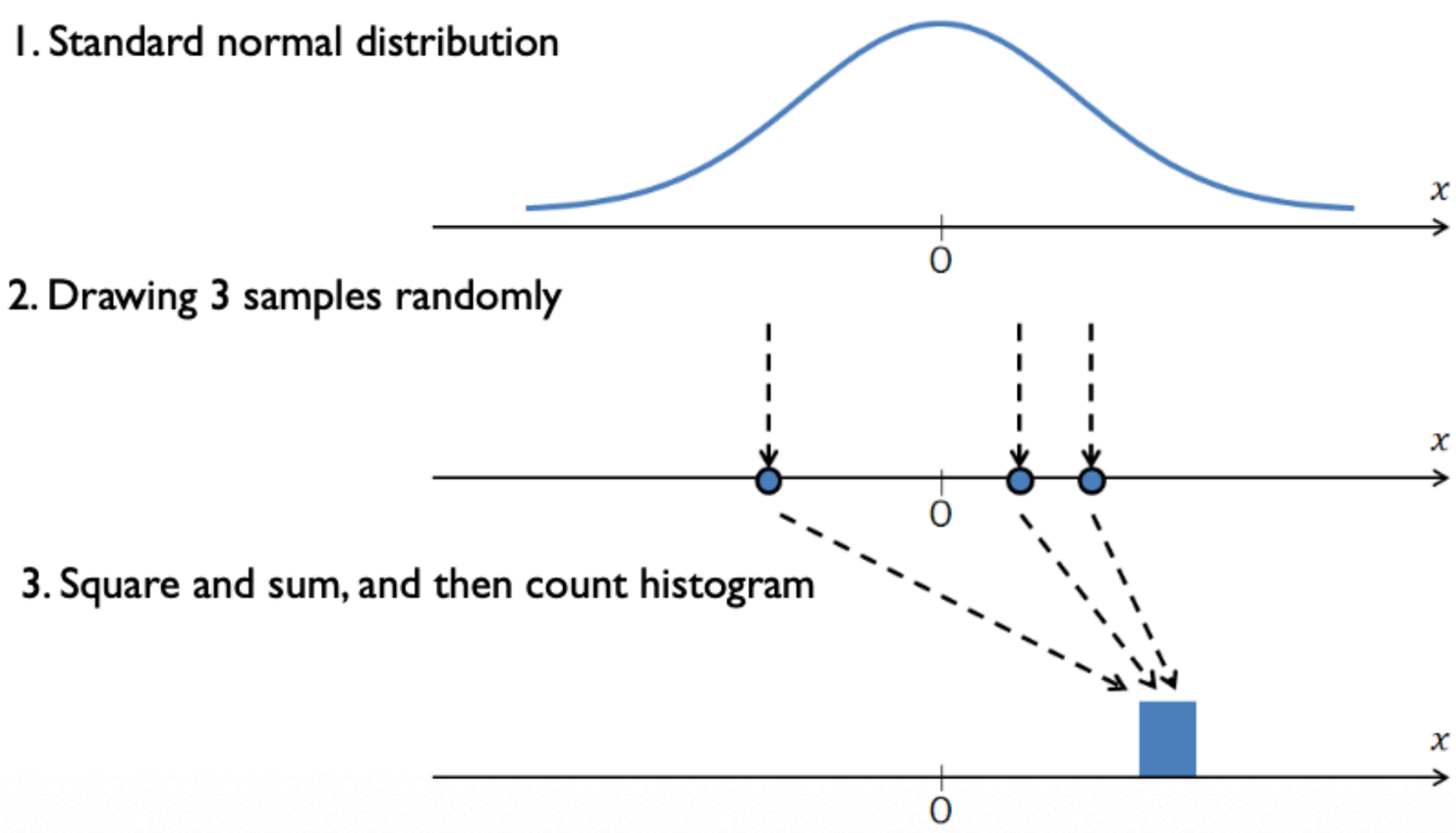
When $df = 1, \chi^2$-distribution

$Z \sim N(0,1) \quad \Rightarrow \quad Z^2 \sim \chi^2(1)$



When $df = 3, \chi^2$-distribution

$$Z_1^2 + Z_2^2 + Z_3^2 \sim \chi^2(3)$$



χ^2-distribution

- If $Z_1, Z_2, \ldots, Z_\nu \overset{i.i.d.}{\sim} N(0,1)$, then the sum of their squares;

$$Q = \sum_{i=1}^{\nu} Z_i^2,$$

is distributed according the χ^2 distribution with ν degrees of freedom.

- This is usually denoted as;

$$Q \sim \chi^2(\nu) \text{ or } Q \sim \chi^2_\nu$$

Properties of the χ^2-distribution

- 1. It is not symmetric.
- 2. The shape of the chi-square distribution depends upon the degrees of freedom, just like Student's *t*-distribution.
- 3. As the number of degrees of freedom increases, the chi-square distribution becomes more symmetric.
- 4. The values are non-negative. That is, the χ^2 values of are greater than or equal to 0.

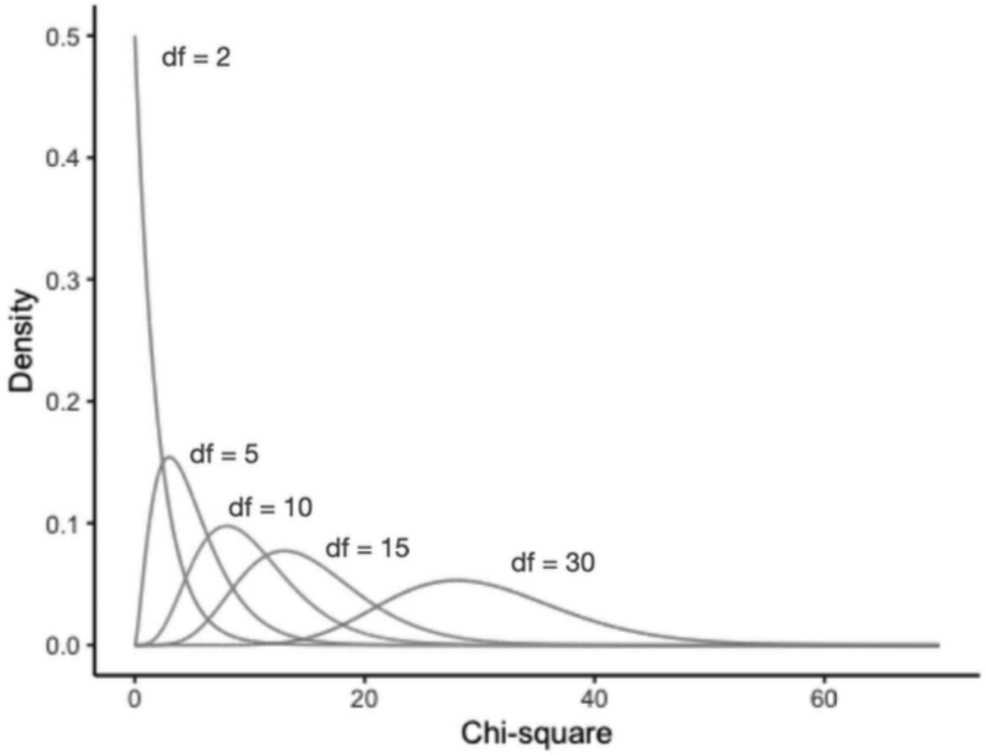


그림 5.1 자유도에 따른 χ^2 분포

2 main Chi-square techniques

The **Chi-square test for goodness of fit** (also called a single sample Chi-square test)

- To test hypotheses about the shape or proportions of a population distribution

The **Chi-square test for independence** (also called contingency table analysis)

- To test whether or not there is a relationship between the two variables

The Chi-Square Test for Goodness-of-Fit

- To test hypotheses about the shape or proportions of a population distribution and determine how well an observed frequency distribution fits an expected pattern of frequencies
- The null hypothesis specifies the proportion of the population that should be in each category.
- The data, called **observed frequencies**, simply count how many individuals from the sample are in each category.
- The proportions from the null hypothesis are used to compute **expected frequencies** that describe how the sample would appear if it were in perfect agreement with the null hypothesis.

Basic Chi-square calculation

1. Set the hypotheses.

$H_0 : O_i = E_i$

2. Compute test statistics.

$$\chi^2_{df} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

O_i: Observed frequency

E_i: Expected frequency

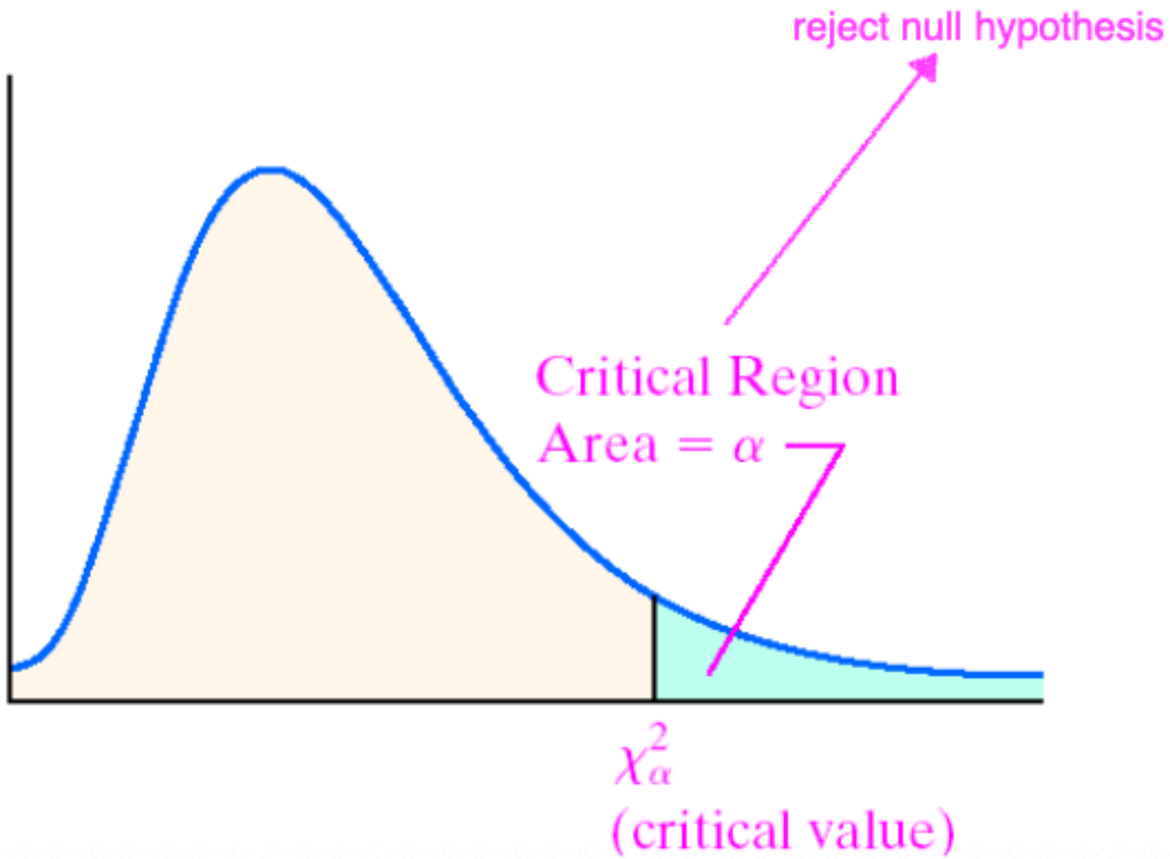
k: Number of categories

3. Compare the critical value to the test statistics.

$\chi^2_{\text{obs}} > \chi^2_{\text{crit}} \Rightarrow$ Reject H_0

4. State the conclusion.

Using χ^2–Table



Exercise: Beer again

- The owner of a tavern wanted to know what if there were any preferences in the types of beer he sold. He recorded the preferences of a random sample of 100 customers and found the following:



10



20



55



15

Exercise: Beer again

1. State H_0 and H_A

- H_0: The population of customers show no preference in buying any one of the four types of beer over the others.
- H_A: In the population of customers, one or more of these types is preferred by customers.

2. Compute the expected occurrence of the events under the null hypothesis

	Heineken	Guiness	Tsingtao	Hoegaarden
Observed	10	20	55	15
Expected	25	25	25	25

Exercise: Beer again

3. Calculate the Chi-square test statistic

$$\chi^2_{df} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(10-25)^2}{25} + \frac{(20-25)^2}{25} + \frac{(55-25)^2}{25} + \frac{(15-25)^2}{25} = 50$$

$$df = C - 1 = 4 - 1 = 3$$

Exercise: Beer again

4. Find the critical value from the chi-square table.

Chi-Square Distribution Table

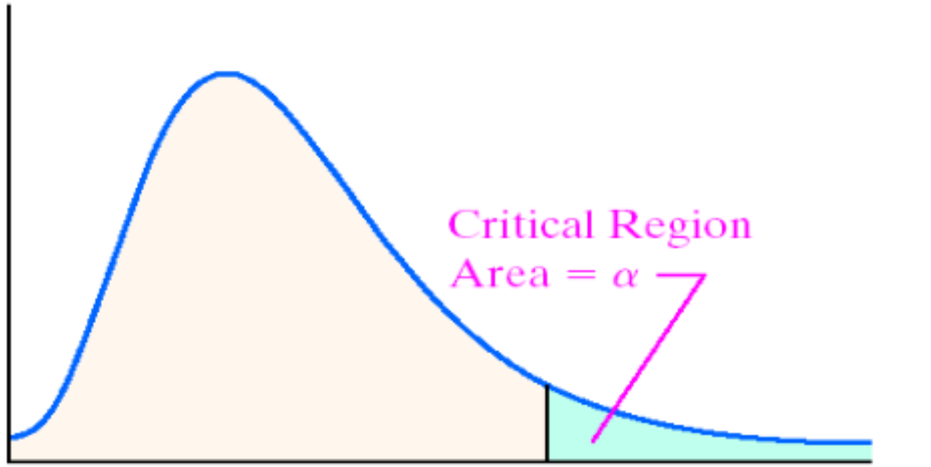
df	Level of Significance										
	0.995	0.975	0.2	0.1	0.05	0.025	0.02	0.01	0.005	0.002	0.001
1	0.00004	0.00098	1.642	2.706	3.841	5.024	5.412	6.635	7.879	9.55	10.83
2	0.01	0.0506	3.219	4.605	5.991	7.378	7.824	9.21	10.6	12.43	13.82
3	0.0717	0.216	4.642	6.251	7.815	9.348	9.837	11.35	12.84	14.8	16.27
4	0.207	0.484	5.989	7.779	9.488	11.14	11.67	13.28	14.86	16.92	18.47
5	0.412	0.831	7.289	9.236	11.07	12.83	13.39	15.09	16.75	18.91	20.52
6	0.676	1.237	8.558	10.65	12.59	14.45	15.03	16.81	18.55	20.79	22.46

Image source: https://itfeature.com/statistical-tables/chi-square-distribution-chi2/#goog_fullscreen_ad

Exercise: Beer again

5. Compare the calculated and critical values & Make a decision

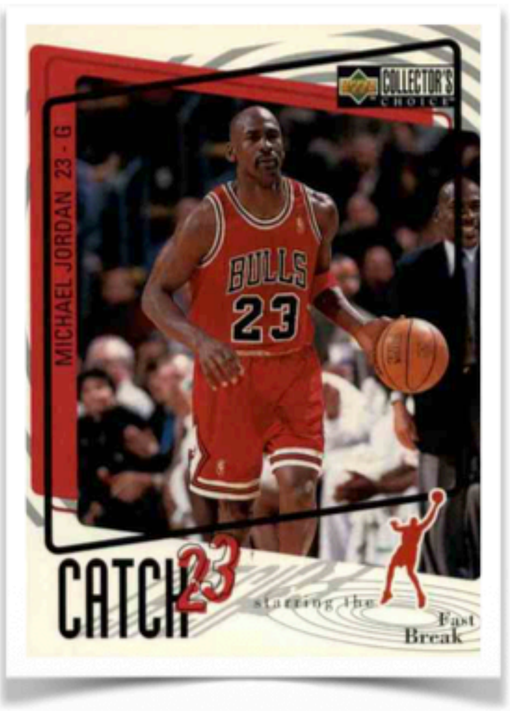
- If $df = 3$, critical value for rejecting H_0 is 7.82. ($\alpha = 0.5$)
- We reject null hypothesis because calculated χ^2 value (50) > 7.82.

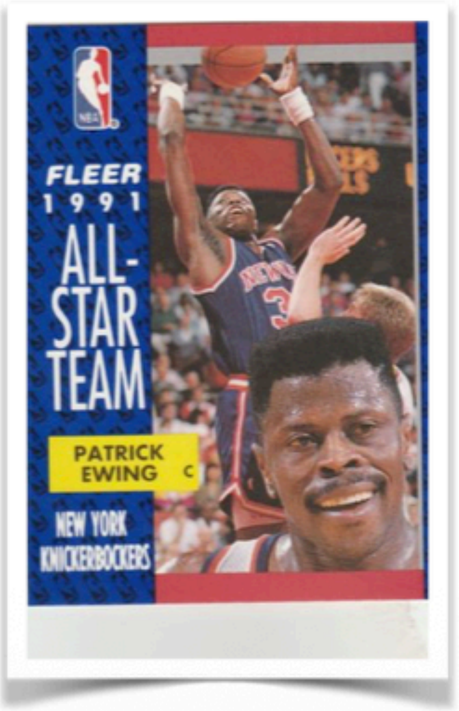


Exercise: Baseball cards

- Acme Toy Company prints basketball cards. The company claims that 30% of the cards are rookies, 60% veterans, and 10% are All-Stars. Suppose a random sample of 100 cards has **50 rookies, 45 veterans, and 5 All-Stars.** Is this consistent with Acme's claim? Test this by using a 0.05 level of significance.







Exercise: Baseball cards

1. State H_0 and H_A

- H_0: The distribution of the observed number of basketball cards ***does not differ*** significantly from the distribution of their proportion that company claimed.
- H_A: The distribution of the observed number of basketball cards ***differ*** significantly from the distribution of their proportion that company claimed.

2. Compute the expected occurrence of the events under the null hypothesis

	Rookie	Veteran	All-Star
Observed	50	45	5
Expected	30	60	10

Exercise: Baseball cards

3. Calculate the Chi-square test statistic

$$\chi^2_{df} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(50-30)^2}{30} + \frac{(45-60)^2}{60} + \frac{(5-10)^2}{10} = 19.58$$

$$df = C - 1 = 3 - 1 = 2$$

Exercise: Baseball cards

4. Find the critical value from the chi-square table.

Chi-Square Distribution Table

df	Level of Significance										
	0.995	0.975	0.2	0.1	0.05	0.025	0.02	0.01	0.005	0.002	0.001
1	0.00004	0.00098	1.642	2.706	3.841	5.024	5.412	6.635	7.879	9.55	10.83
2	0.01	0.0506	3.219	4.605	5.991	7.378	7.824	9.21	10.6	12.43	13.82
3	0.0717	0.216	4.642	6.251	7.815	9.348	9.837	11.35	12.84	14.8	16.27

5. Compare the calculated and critical values & Make a decision

- $\chi^2_{\text{obs}} = 19.58 > \chi^2_{\text{crit}} = 5.99 \quad \& \quad P\left(\chi^2_2 \geq 19.58\right) < 0.0001 \quad \Rightarrow \quad$ Reject H_0
- Since the *p*-value (0.0001) is less than the significance level (0.05), we can reject the null hypothesis. Therefore, The distribution of the observed number of basketball cards significantly ***differ*** from the distribution of their proportion that company claimed.

The χ^2 Test for Independence (Contingency table χ^2 test)

The second chi-square test, *the chi-square test for independence,* can be used and interpreted in two different ways:

- 1. Testing hypotheses about the relationship between two variables in a population, or
- 2. Testing hypotheses about differences between proportions for two or more populations.

Example: Job performance and Self-esteem

- A researcher is examining the relationship between job performance and self-esteem. A sample of $n = 150$ is selected. Each subject is classified by level of job performance and level of self-esteem. The data are as follows:

		Self-esteem			
		High	Medium	Low	
Job performance	High	17	32	11	60
	Low	13	43	34	90
		30	75	45	150

Example: Job performance and Self-esteem

Expected values for cells

- Calculate marginal totals first
- Then calculate expected cell values: $(Total_{row} * Total_{column})/GrandTotal$
- Repeat process for each cell
- Cross-check by:
  - Summing expected values across all cells (should sum to the observed grand total)
  - Summing marginal totals (should reproduce observed marginal totals)

Example: Job performance and Self-esteem

Determine the expected frequencies (Red numbers indicate expected frequencies)

		Self-esteem			
		High	Medium	Low	
Job performance	High	17 60*30/150 = 12	32 60*75/150 = 30	11 60*45/150 = 18	60
	Low	13 90*30/150 = 18	43 90*75/150 = 45	34 90*45/150 = 27	90
		30	75	45	150

*We call the blue cells (the totals across rows and columns) ***marginal totals***, and the red cell the ***grand total.***

Example: Job performance and Self-esteem

3. Calculate the Chi-square test statistic

$$\chi^2_{df} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(17-12)^2}{12} + \frac{(32-30)^2}{30} + \frac{(11-18)^2}{18} + \frac{(13-18)^2}{18} + \frac{(43-45)^2}{45} + \frac{(34-27)^2}{27} \approx 8.23$$

$$df = (R-1)(C-1) = (2-1)(3-1) = 2$$

4. Compare the calculated and critical values & Make a decision

- Because calculated χ^2 value 8.23 > critical χ^2 value for rejecting H_0 5.99, we reject null hypothesis.
- There is a statical significant relationship between job performance and self-esteem.

Effect Sizes for Chi-Square Tests

- A significant chi-square test indicates that observed frequencies differ from expected frequencies.
- Effect size measures the magnitude of this discrepancy or the strength of association between categorical variables.
- Common effect size measures include:
  - Goodness-of-Fit test → Cohen's *w*
  - Independent test
    - 2x2 → Phi coefficient (ϕ)
    - RxC → Cramér's *V*, Tschuprow's *T*

Effect Sizes for Chi-Square Tests

- When both variables in the chi-square test for independence consist of exactly two categories (the data form a 2x2 matrix), it is possible to re-code the categories as 0 and 1 for each variable and then compute a correlation known as a **phi-coefficient** that measures the strength of the relationship.

- The value of the phi-coefficient, or the squared value which is equivalent to an R^2, is used to measure the effect size.

- When there are more than two categories for one (or both) of the variables, then you can measure effect size using a modified version of the phi-coefficient known as Cramer's V.

- The value of V is evaluated much the same as a correlation.

Cohen's w and Phi Coefficient

- Cohen's w measures the discrepancy between observed and expected proportions.
- Phi (ϕ) measures the association between two binary variables.
  - For a 2×2 contingency table, Cohen's w, ϕ, and Cramér's V are equivalent: $w = |\phi| = V$

	Y = 1	Y = 0	Total
X = 1	a	b	a+b
X = 0	c	d	c+d
Total	a+c	b+d	N

$$w = \sqrt{\frac{1}{N}\sum_{i=1}^{k}\frac{(O_i - E_i)^2}{E_i}} = \sqrt{\frac{\chi^2}{N}}$$

$$\phi = \frac{ad - bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}$$

$w = .10$ **Small effect size**
$w = .30$ **Medium effect size**
$w = .50$ **Large effect size**

Cramer's V

- Cramér's V measures the strength of association between two categorical variables.
- It can be applied to $R \times C$ contingency tables.
- Values range from 0 (no association) to 1 (maximum association).
- If $df^* > 5$, convert Cramer's V to Cohen w to interpret effect size (Cohen, 1988): $w = V\sqrt{\min(r-1, c-1)}$

$$V = \sqrt{\frac{\chi^2}{N \min(r-1, c-1)}}$$

- r: number of rows
- c: number of columns
- N: sample size

표 5.4 Cramér's V에 대한 해석 (Cohen, 1988)

df^*	negligible	small	medium	large
1	$0 < .10$	$.10 < .30$	$.30 < .50$	$\geq .50$
2	$0 < .07$	$.07 < .21$	$.21 < .35$	$\geq .35$
3	$0 < .06$	$.06 < .17$	$.17 < .29$	$\geq .29$
4	$0 < .05$	$.05 < .15$	$.15 < .25$	$\geq .25$
5	$0 < .05$	$.05 < .13$	$.13 < .22$	$\geq .22$

Tschuprow's T

- Tschuprow's T is another measure of association between categorical variables.
- Unlike Cramér's V, it accounts symmetrically for both table dimensions.
- $T = V$ for square contingency tables.
- $T \leq V$ for rectangular contingency tables.

$$T = \sqrt{\frac{\chi^2}{N\sqrt{(r-1)(c-1)}}}$$

- r: number of rows
- c: number of columns
- N: sample size

Comparison of Cramér's V and Tschuprow's T

$2 \times 2: \quad T = V = |\phi|$

$3 \times 3: \quad T = V$

$2 \times 4: \quad T < V$ *(when association is nonzero)*

Assumptions for Chi-square Tests on Contingency Tables

Independence assumption

- Individual observations must be independent of all other observations.
- Independent means that there should be one observation per observation (or subject), which means that observations in each group (level) should not be correlated with each other.
- In addition, individual observations must exist in only one cell, meaning that each observation must not be paired in any way. For example, it is not allowed for a single observation to have pre- and post-measurements over time.

Assumptions for Chi-square Tests on Contingency Tables

Minimum expectation

- "Since χ^2-test has been established as the limiting distribution of χ^2 in large samples," there are working rules that the smallest expected number in any class should be recommended (Cochran, 1952, 1954).

- For tables larger than 2x2: "No more than 20% of the expected counts are less than 5, and all individual expected counts are 1 or greater" (Yates, Moore & McCabe, 1999, p. 734).

- For 2x2 Tables (Cochran, 1952, 1954): All expected counts should be 10 or greater. If any expected counts are less than 10, but greater than or equal to 5.

Assumptions for Chi-square Tests on Contingency Tables

Cochran's recommendation (Cochran, 1954)

- 1. 정규분포 또는 포아송 분포와 같은 단봉분포(unimodal distributions)의 χ^2 적합도 검정의 경우 최소 기대빈도는 1 이상이 되어야 한다.
- 2. 2x2 분할표의 총 표본크기 N이 20 미만이거나, $20 < N < 40$이면서 가장 작은 기대빈도가 5 미만이면 Fisher's exact test를 사용한다. 만일 $N > 40$인 경우에는 Yates' correction for continuity χ^2-test를 사용한다.
- 3. 자유도가 1보다 큰 분할표의 경우, 만일 5보다 작은 기대빈도가 전체 셀의 20%를 넘는다면(예: 5개 셀 중 1개 이상 혹은 10개 셀 중 2개 셀 이상) Fisher's exact test를 사용한다.

Assumptions for Chi-square Tests on Contingency Tables

Yates' Correction

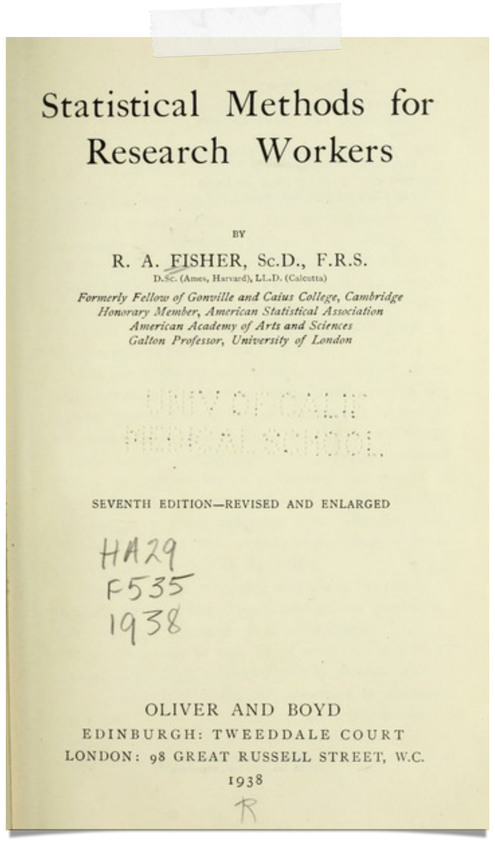
- some authors suggest that Yates' Correction for continuity should be applied.

$$\chi^2_{Yates} = \sum_{i=1}^{N} \frac{(|O_i - E_i| - 0.5)^2}{E_i}$$

- However, the use of Yates' correction is controversial, and is not recommended by all scholars.

- If any expected counts are less than 5, then some other test should be used (e.g., Fisher exact test for 2x2 contingency tables).

Fisher’s exact test


Statistical Methods for Research Workers

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Sir Ronald Aylmer, F. (1934). *Statistical methods for research workers, 5th edition.* Oliver and Boyd.

- The Fisher Exact test is a test of significance that is used in cases of **small samples.**

- It is hard to say how many samples are small, but in general, it is better to use a Fisher's exact test than a Chi-square test when…
  - Total number of sample (N) < 20 with a 2x2 matrix, or
  - $N >$ 20, but expected cell frequency is 5 or greater is less than 80% of cells

Fisher’s exact test

	Men	Women	*Row Total*
Studying	***a***	***b***	*a + b*
Non-studying	***c***	***d***	*c + d*
Column Total	*a + c*	*b + d*	*a + b + c + d (=n)*

$$p = \frac{\binom{a+b}{a}\binom{c+d}{c}}{\binom{n}{a+c}} = \frac{\binom{a+b}{b}\binom{c+d}{d}}{\binom{n}{b+d}} = \frac{(a+b)!\,(c+d)!\,(a+c)!\,(b+d)!}{a!\;\;b!\;\;c!\;\;d!\;\;n!}$$

$$p = \binom{10}{1}\binom{14}{11}/\binom{24}{12} = \frac{10!\;14!\;12!\;12!}{1!\;9!\;11!\;3!\;24!} \approx 0.001346076$$

Example source: https://en.wikipedia.org/wiki/Fisher's_exact_test